

Week 1 Article

Now that you've completed your first week of learning Excel as part of your Data Science & Analytics course, it's time to reflect and share what you've learned with the world. Writing articles builds your online portfolio as a future data analyst, helps you retain what you've learned and prepares you for real-world communication as a data professional.

Write and publish an article on DEV.to (or your preferred blog site).

You will write your first blog article to explain how Excel is used in the real world, based on your current understanding and what you've learned this week.

Topic:

"How Excel is Used in Real-World Data Analysis"

Length: 300–500

words

Platform:

Create and publish your article on <https://dev.to> If you don't have an account, follow these steps:

1. Go to <https://dev.to>
2. Click "Create Account"
3. Use your email or GitHub to sign up
4. Complete your profile with a short bio

What to Include in Your Article

- A brief introduction to what Excel is
- At least 3 ways Excel is used in real-world data analysis (e.g., business decisions, financial reporting, marketing performance, etc.)
- Mention at least 3 Excel features or formulas you've learned so far and explain how they can be applied
- Your personal reflection:
How has learning Excel changed the way you see data?

Submission

1. Links with published article should be sent to datascience@luxdevhq.com by Sunday 7th June 2026 4pm

Data Analysis Dashboard Project: Jumia Product Analysis

Project Title: "Jumia Product Performance Dashboard: Analyzing Pricing, Discounts, and Customer Reviews"

Objective:

The goal of this project is to create an interactive Excel dashboard that provides insights into the performance of products listed on Jumia. Students will analyze the dataset to understand the relationship between pricing, discounts, customer reviews, and ratings. The dashboard will help Jumia identify trends, popular products, and areas for improvement.

Dataset Overview:

The dataset contains the following columns:

- **Product:** Name of the product.
- **Current Price:** The current selling price of the product (in KSh).
- **Old Price:** The original price of the product (in KSh).
- **Discount:** The discount percentage offered on the product.
- **Review:** The number of reviews for the product.
- **Rating:** The average customer rating of the product (out of 5).

Project Tasks:

Students will work in groups to analyze the dataset and create an interactive Excel dashboard. The project will be divided into the following steps:

Step 1: Data Cleaning and Preparation

1. Data Cleaning:

- Check for missing values in the dataset (e.g., missing reviews or ratings). ○

Remove duplicates if any.

- Ensure all price columns are in numeric format (remove "KSh" and commas).
- Standardize the rating column to a numeric format (e.g., convert "4.5 out of 5" to 4.5).

2. Data Enrichment:

- Calculate the absolute discount amount for each product (Old Price - Current Price).
- Create a new column to categorize products based on their rating (e.g., "Poor" for ratings below 3, "Average" for 3-4, and "Excellent" for 4.5 and above).
- Create a new column to categorize products based on discount percentage (e.g., "Low Discount" for <20%, "Medium Discount" for 20-40%, "High Discount" for >40%).

Step 2: Data Analysis

1. Descriptive Statistics:

- Calculate the average current price, old price, discount percentage, and rating across all products.
- Identify the most expensive and least expensive products.
- Determine the average discount percentage and rating for each product category.

2. Trend Analysis:

- Analyze the relationship between discount percentage and number of reviews. Do higher discounts lead to more reviews?
- Analyze the relationship between rating and number of reviews. Do higher-rated products have more reviews?
- Identify the top 5 products with the highest ratings and the top 5 products with the lowest ratings.

3. Product Performance:

- Identify the top 10 products with the highest discounts. ○ Identify the top 10 products with the most reviews.
- Compare the performance of products with high discounts versus low discounts (e.g., average rating, number of reviews).

Step 3: Dashboard Design

1. Dashboard Layout:

- Create a visually appealing and interactive dashboard in Excel. ○ Use Pivot Tables, Charts, and Slicers to make the dashboard interactive.
- Include the following sections in the dashboard:
 - Overview: Key metrics such as total number of products, average rating, average discount percentage, and total reviews.

- **Product Performance:** A table or chart showing the top 10 products by rating, reviews, and discounts.
- **Trend Analysis:** Charts showing the relationship between discount percentage and reviews, and between rating and reviews.
- **Product Categories:** A breakdown of products by rating category (Poor, Average, Excellent) and discount category (Low, Medium, High).

2. Visualizations:

- Use bar charts, line charts, and pie charts to visualize trends and comparisons.
- Use conditional formatting to highlight key metrics (e.g., high discounts, low ratings).
- Add slicers for filtering data by rating, discount category, and product type.

Example Insights to Explore:

- Are higher discounts leading to more sales (as indicated by the number of reviews)?
- Do higher-rated products have higher prices, or are they more affordable?
- Which product categories (e.g., home appliances, tools, decorations) have the highest ratings and discounts?
- Are there any products with high discounts but low ratings? What could be the reason?

Tools Required:

- Microsoft Excel (Pivot Tables, Charts, Slicers, Conditional Formatting)
- Basic knowledge of data analysis and visualization techniques.

Deliverables:

A cleaned and enriched Excel dataset.

2. A neat interactive Excel dashboard with visualizations and slicers.
3. All assignments (github links containing the excel workbook) should be submitted to datascience@luxdevhq.com by 4PM Friday 12th June 2026.